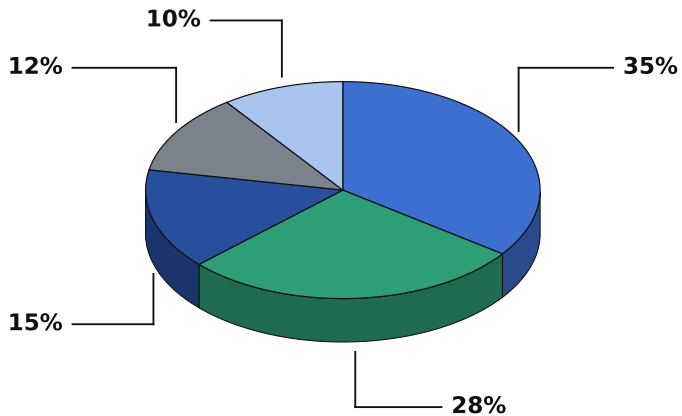
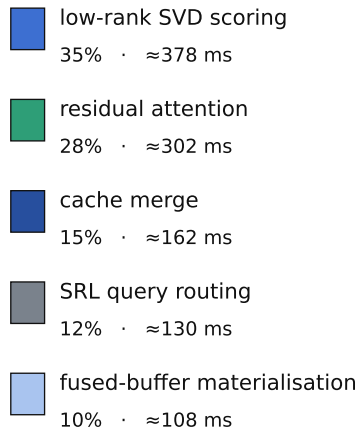




one decode step $\approx 1,080$ ms (instrumented, 16k)



63% of the step is the per-token low-rank contraction $q \cdot V_K$ plus the residual attend — not buffer materialisation. Collapsing the two into one fused launch is the open optimisation.